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JANUARY 2010

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

PHYSICS

Paper 02 – General Proficiency

$2\frac{1}{2}$ hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **SIX** questions.

2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.

3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.

4. All working **MUST** be **CLEARLY** shown.

5. The use of non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.

6. Mathematical tables are provided.

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Graph paper for Question 1

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SECTION A

Attempt ALL questions.

You MUST write your answers in this answer booklet.

1. In an experiment to investigate Charles' Law, a Physics student used a fixed mass of dry air trapped in a glass capillary tube as seen in Figure 1. The student recorded in Table 1 how the length of the air column varied with temperature.

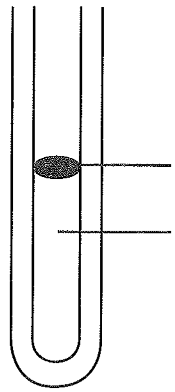

Figure 1. Diagram showing air trapped in a capillary tube

TABLE 1: RESULTS OF THE EXPERIMENT

Length of the air column, L/mm	152.0	158.0	163.0	170.0	179.0	182.0
Temperature, T/°C	14.0	29.0	40.0	57.5	78.0	85.0
Temperature, T / K						

(a) Complete Table 1 by calculating the temperature, T / K. **(3 marks)**

(b) Use the readings from the completed Table 1 to plot a graph of length of the air column (L/mm) against Temperature (T/K) on the graph page on page 2.

The L-axis starts at 140 mm and the T-axis starts at 270 K. **(6 marks)**

(c) From your graph, calculate the slope, S.

(**4 marks)**

(d) What information does the slope (gradient) of the graph provide?

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(1 mark)

(e) Use your graph to determine the value of the length of the air column at 273 K.

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(1 mark)

(f) State Charles' law.

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(4 marks)

(g) In a related practical activity, 2 litres of a gas were heated from 35 $^{\circ}$C to 75 $^{\circ}$C. If the pressure was kept constant, calculate the final volume of the gas.

(6 marks)

Total 25 marks

2. (a) (i) Complete the following table, by inserting the correct quantity, formula and unit.

Quantity	**Formula**	**Unit**
	F = ma	
Potential Energy		
		$kgms^{-1}$

(3 marks)

(ii) State the 'law of conservation of linear momentum'.

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(4 marks)

(b) (i) <u>"BIG CRASH ON THE HIGHWAY"</u>

"Two trucks of equal mass collided head-on at the same speed on the busy East-West Highway. They both remained stationary on impact."

Explain this crash in terms of conservation of linear momentum. Assume the masses of other contents of the trucks are equal.

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(4 marks)

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(ii) A police recruit, while training, shot at a stationary target of mass 5.0 kg, with a bullet of mass 0.1 kg. The target was mounted on low-fiction wheels and as soon as the bullet struck the target, the target with the embedded bullet sped off with a velocity of 6.0 m s^{-1}.

Calculate the velocity of the bullet just before it hit the stationary target.

(4 marks)

Total 15 marks

3. (a) Two types of waves are longitudinal and transverse waves.

What is meant by the term 'longitudinal wave'?

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(2 marks)

(b) Figure 2 shows a wavetrain in which the points A to I and *w* to *z* are shown.

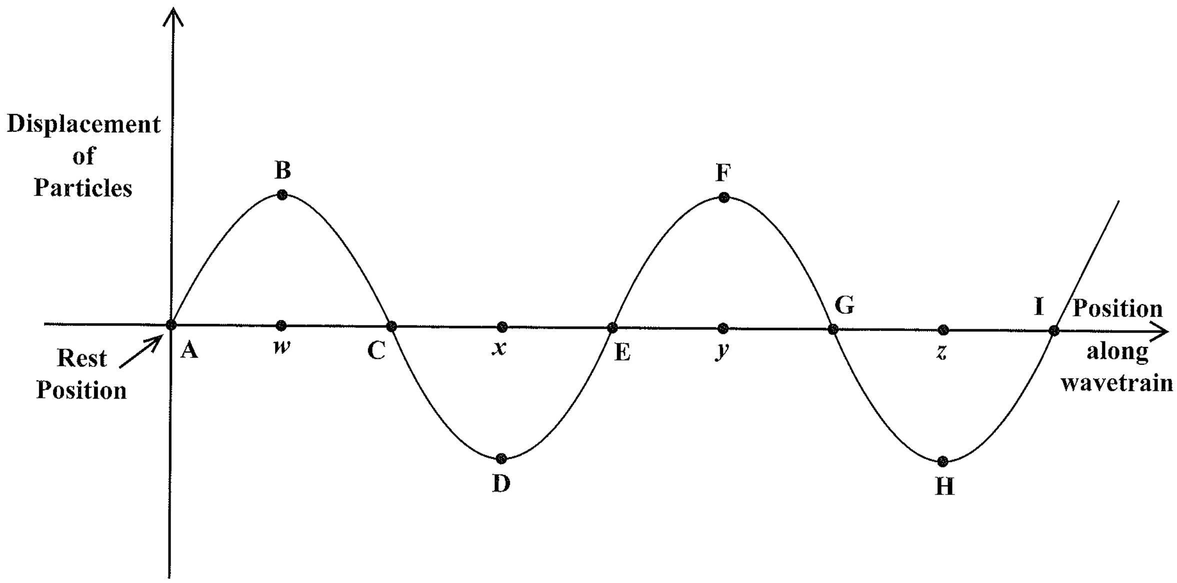

Figure 2. A wavetrain

Identify, using TWO letters, the distance between any TWO points of the following in the wavetrain:

(i) A wavelength ______________________________

(ii) The amplitude of the wave ______________________________

(2 marks)

(c) A slinky spring can be used to generate a transverse waveform. In the box below, draw and label this waveform, clearly identifying

- (i) particle movement
- (ii) waveform movement
- (iii) at least one crest and one trough.

(3 marks)

(d) (i) A steelpan produces a sound of frequency 0.350 KHz. The speed of sound in air is 340 m s^{-1}. What is the wavelength of the sound generated by the pan?

(4 marks)

(ii) If, this sound is generated under water instead, what is the frequency that would be detected?

(1 mark)

(iii) The sound wave generated under water is refracted at the water - air boundary. Calculate the refractive index of water, given that the wavelength of this sound in water is 1.29 m.

(3 marks)

Total 15 marks

SECTION B

Attempt ALL questions.

You MUST write your answers in the space provided after each question.

4. (a) A young student has just entered primary school. She is accustomed to having her home-cooked meal warmed up at lunch time in the microwave oven at her pre-school. She misses this and suggests to her parents that they buy her a flask similar to the one shown in Figure 3.

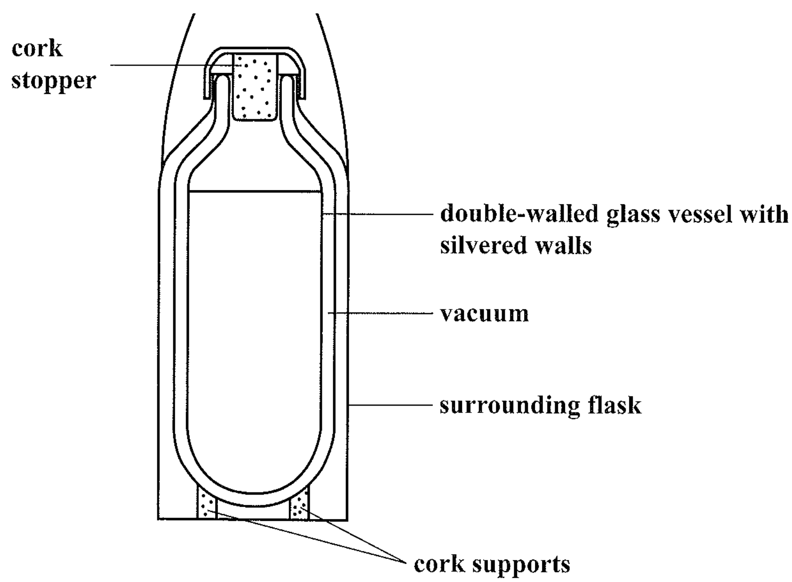

Figure 3. Cross-section of a vacuum flask

Using the information in Figure 3, explain how the home-cooked meal is kept warm.

(6 marks)

(b) A solar water heater system consists of a solar collector of area 5 m^2, a storage tank and hidden connecting pipes as shown in Figure 4.

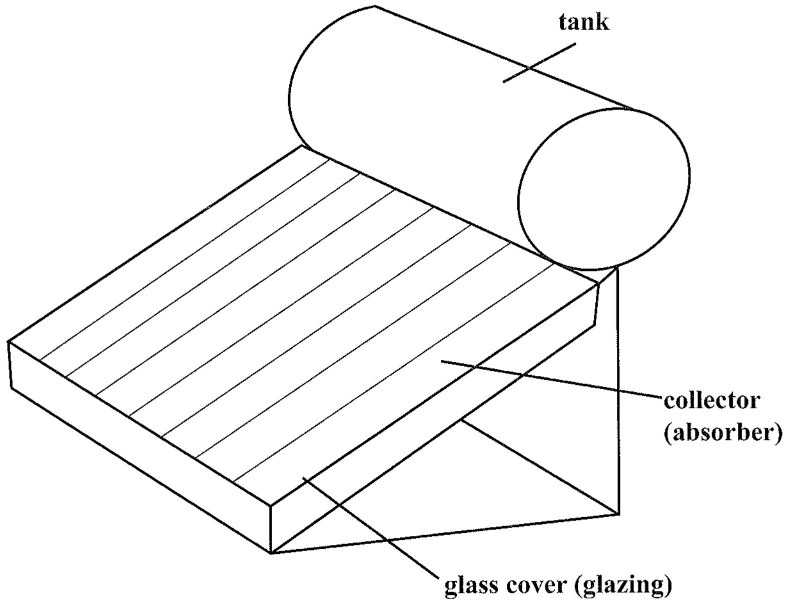

Figure 4. Diagram showing a solar water heater

Assume that the average solar radiation incident on Castries, St. Lucia is 5 $kWhm^{-2}$ per day, and the collector absorbs 95% of the solar radiation incident on it:

(i) Calculate the energy in kWh collected by the absorber per day. **(2 marks)**

(ii) Calculate how much energy per day can be used to heat water if the collector emits 50% of the heat energy it receives from the atmosphere. **(2 marks)**

(iii) A glass cover is now placed on the solar collector. The glass transmits 80% of the sunlight to the collector and allows 8% of the heat back into the atmosphere (the glass house effect). How much energy per day is now available to heat the water? **(2 marks)**

(iv) What mass of water can the glazed collector heat from 25°C to 55°C during the day? **(3 marks)**

1 kWh = 3.6×10^6J
Specific heat capacity of water, C = 4200 J kg^{-1} k^{-1}

Total 15 marks

Write your answer to Question 4 here.

5. (a) With the aid of a CLEARLY labelled circuit diagram, explain how the I – V characteristics of an unknown electrical component, X, may be determined. **(6 marks)**

(b) The lights in a room are connected as shown in Figure 5. Each bulb has a resistance of 1000 Ω.

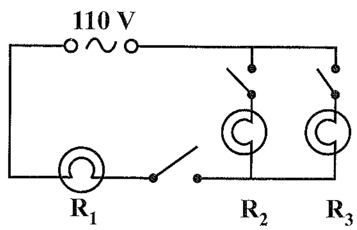

Figure 5. Circuit for lights in a room

Determine

(i) the total resistance

(ii) the current drawn from the supply if all three bulbs are lit. **(6 marks)**

(c) A microwave oven consumes 1100 W when operated on high power. What is the MINIMUM current rating for a fuse to be used to protect the oven if the mains potential difference is 110 V? **(3 marks)**

Total 15 marks

Write your answer to Question 5 here.

6. (a) It is suspected that a certain radioactive source emits alpha, beta and gamma radiations. With the aid of a diagram, describe how the presence of any TWO of the three types of radiation in such sample could be confirmed. **(6 marks)**

(b) Radioactive carbon $\left({}^{14}_{6}\text{C}\right)$ loses a beta particle to become Nitrogen (N).

Write a nuclear equation to represent this nuclear reaction. **(3 marks)**

(c) Calculate the energy released in the solar fusion of deuterium represented by

$${}^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + \text{energy}$$

Assume $\quad u = 1.66 \times 10^{-27}$ kg

${}^{2}_{1}\text{H}$ has an atomic mass of 2.0140 u

${}^{4}_{2}\text{He}$ has an atomic mass of 4.0026 u

(6 marks)

Total 15 marks

Write your answer to Question 6 here.